



requirements.txt
↓
flet 0.22.0

button: onclick senza ()

Graphical Interfaces

Intro to Flet library, MVC Pattern

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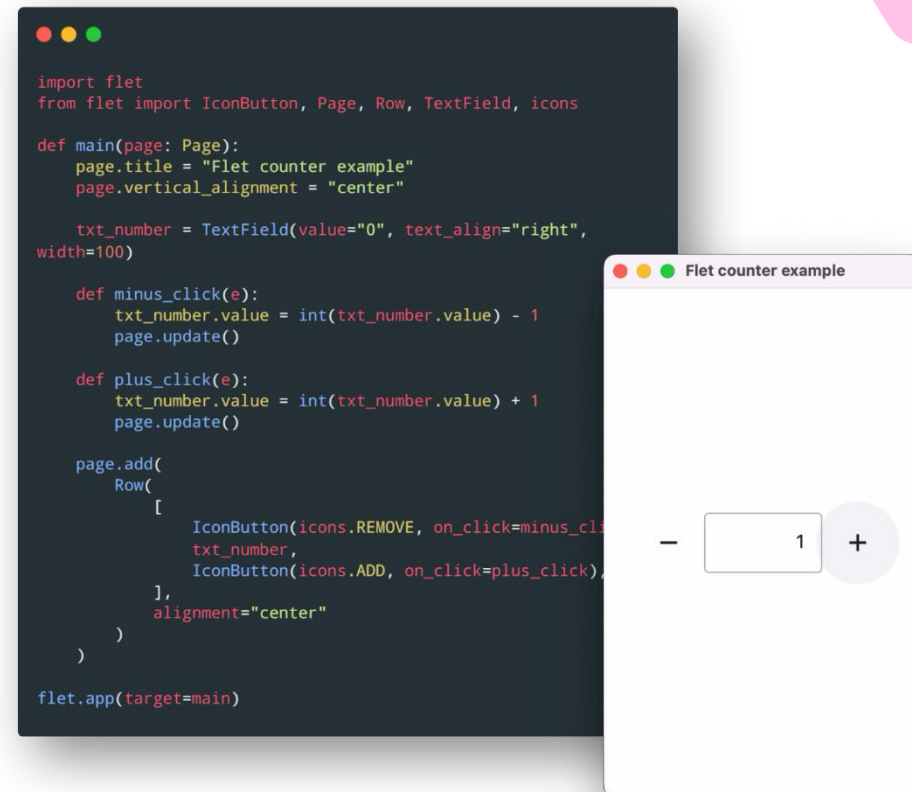
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.text("...")
.row([..., ...])

txtIn → input
txtOut → output



Nel controller, sempre 2 argomenti nelle funzioni (self, e)

Graphical Interfaces

- Why GUIs
- Building GUIs with FLET
 - Intro to flet library
 - Controllers
- Pattern MVC
 - Motivation and logic of Model-view-controller
 - Code skeleton

MODULI :

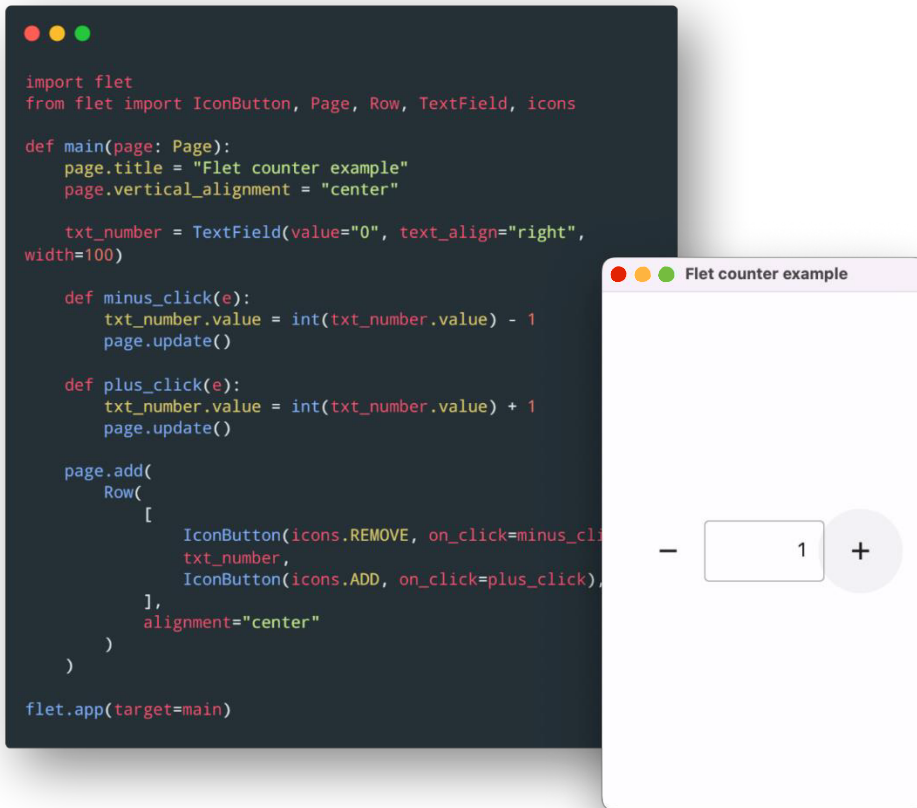
Model → logica

View → interfaccia
grafica

Controller → Model
↕
View

nel modulo
UI, dove si
deve creare 1
file -- init --

Why GUIs



- Enhanced user-friendly interaction
- Productivity boost
- Accessibility and inclusivity in software
- Visual representation of data

GUIs in Python

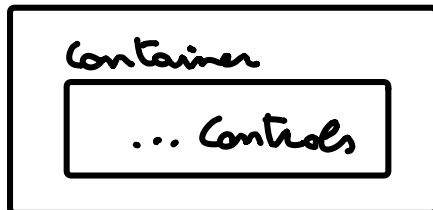
- Few alternatives
 - Tk
 - Default graphic library
 - basic set of widgets for building graphical user interfaces.
 - Flutter
 - Proposed by Google as a multi-platform UI software devel kit
 - Flet
 - Library to build flutter apps in python
 - Easily build realtime web, mobile and desktop apps
 - Often prettier, with minimal frontend knowledge 😊
 - <https://flet.dev>

← NOI

Let's start

- Our program will end with a call to `flet.app()`, where the app starts waiting for new user sessions
- Flet apps are organized in containers, where we put all the graphical elements

Container



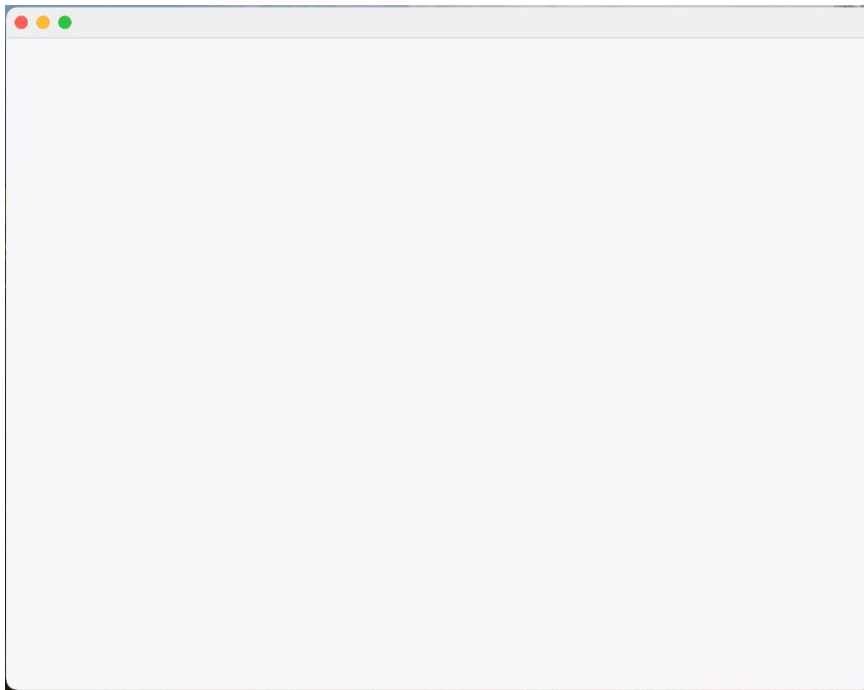
! IMPORTANTE !

```
1 import flet as ft
2
3 def main(page: ft.Page):
4     # add/update controls on Page
5     pass
6
7 ft.app(target=main)
```

Page

Let's start

- The top-most container is View (default option is FLET_APP)

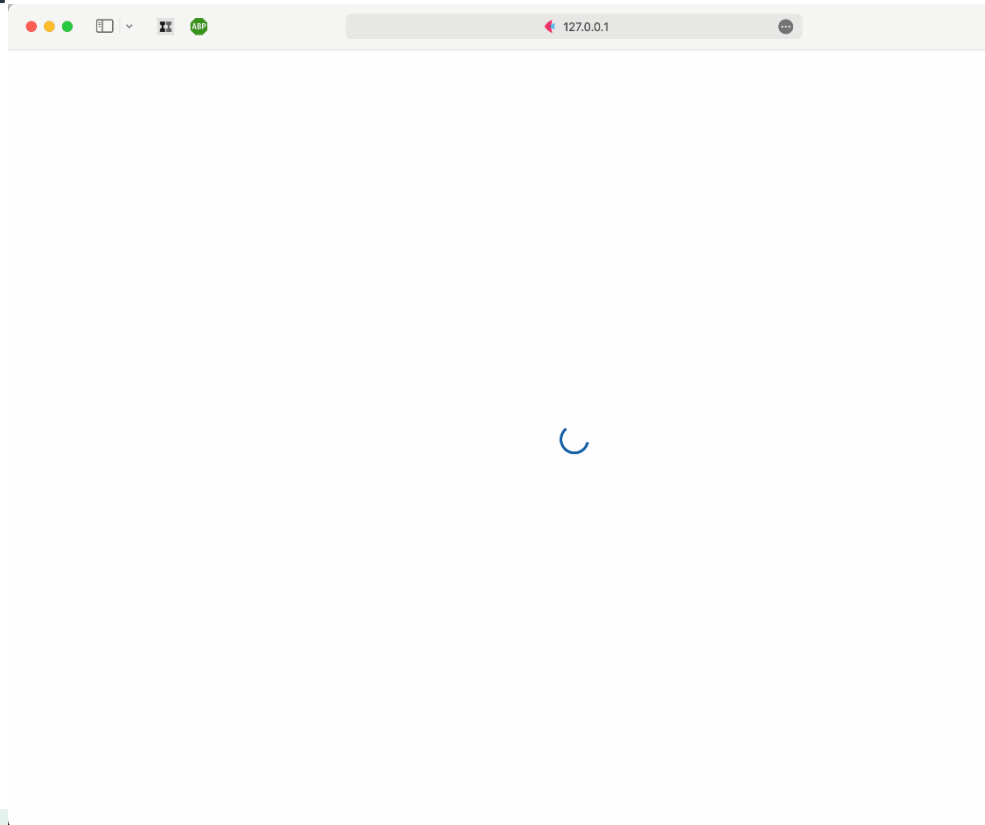


```
1 import flet as ft
2
3 def main(page: ft.Page):
4     # add/update controls on Page
5     pass
6
7 ft.app(target=main, view=ft.AppView.FLET_APP)
```

↓
Container +
grande

Let's start

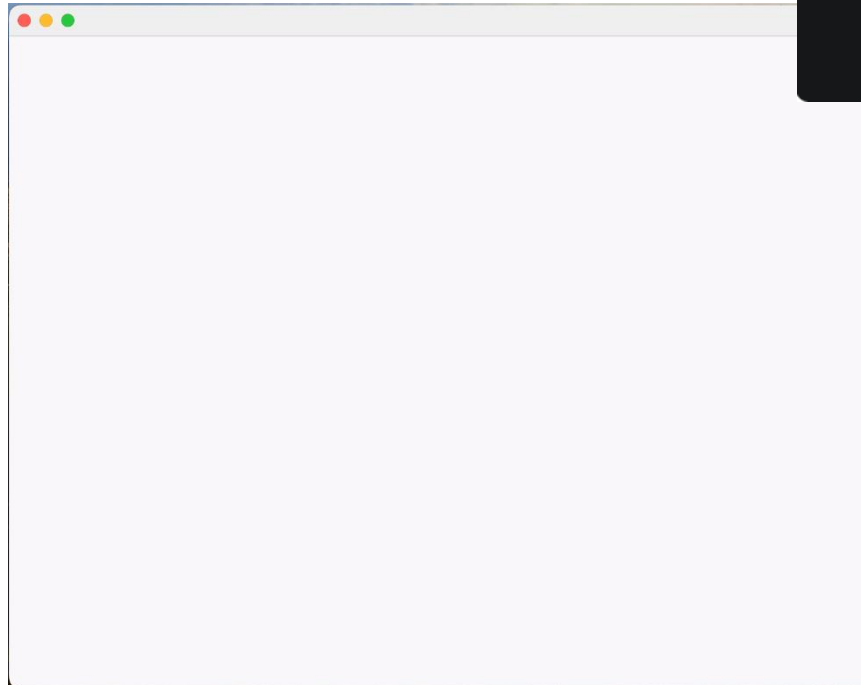
- But we can easily build a browser app



```
1 import flet as ft
2
3 def main(page: ft.Page):
4     # add/update controls on Page
5     pass
6
7 ft.app(target=main, view=ft.AppView.WEB_BROWSER)
```

Let's start

- Within View, we define a Page container, where we will place all the graphic elements.



```
1 import flet as ft
2
3 def main(page: ft.Page):
4     # add/update controls on Page
5     pass
6
7 ft.app(target=main)
```




GUI elements

- GUI elements are called Controls and are simply python classes which need to be instantiated in a Page (root Control)
- For example, to add some text we can use `ft.Text()`

GUI elements

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```
1 import flet as ft
2
3 def main(page: ft.Page):
4     mytext = ft.Text(value="Ciao amici di TdP2024 :-)", color="Blue")
5
6
7 ft.app(target=main)
```

GUI elements

- GUI elements are called Controls and are simply python classes which need to be instantiated in a Page (root Control)
- For example, to add some text we can use `ft.Text()`

```
1 import flet as ft
2
3 def main(page: ft.Page):
4     mytext = ft.Text(value="Ciao amici di TdP2024 :-)", color="Blue")
5     page.controls.append(mytext)
6     page.update()
7
8
9 ft.app(target=main)
```

GUI elements

- GUI elements are called Controls and are simply python classes which need to be instantiated in a Page (root Control)
- Shortcut `page.add()` to directly add the control and update the page.

`.add` → Append
+
Update

```
1 import flet as ft
2
3 def main(page: ft.Page):
4     mytext = ft.Text(value="Ciao amici di TdP2024 :-)", color="Blue")
5     page.controls.append(mytext)
6     page.update() → aggiunge controlli
7
8     page.add(ft.Text(value="Quest'anno in Python")) ↘
9
10
11 ft.app(target=main) update incluso
```

GUI elements

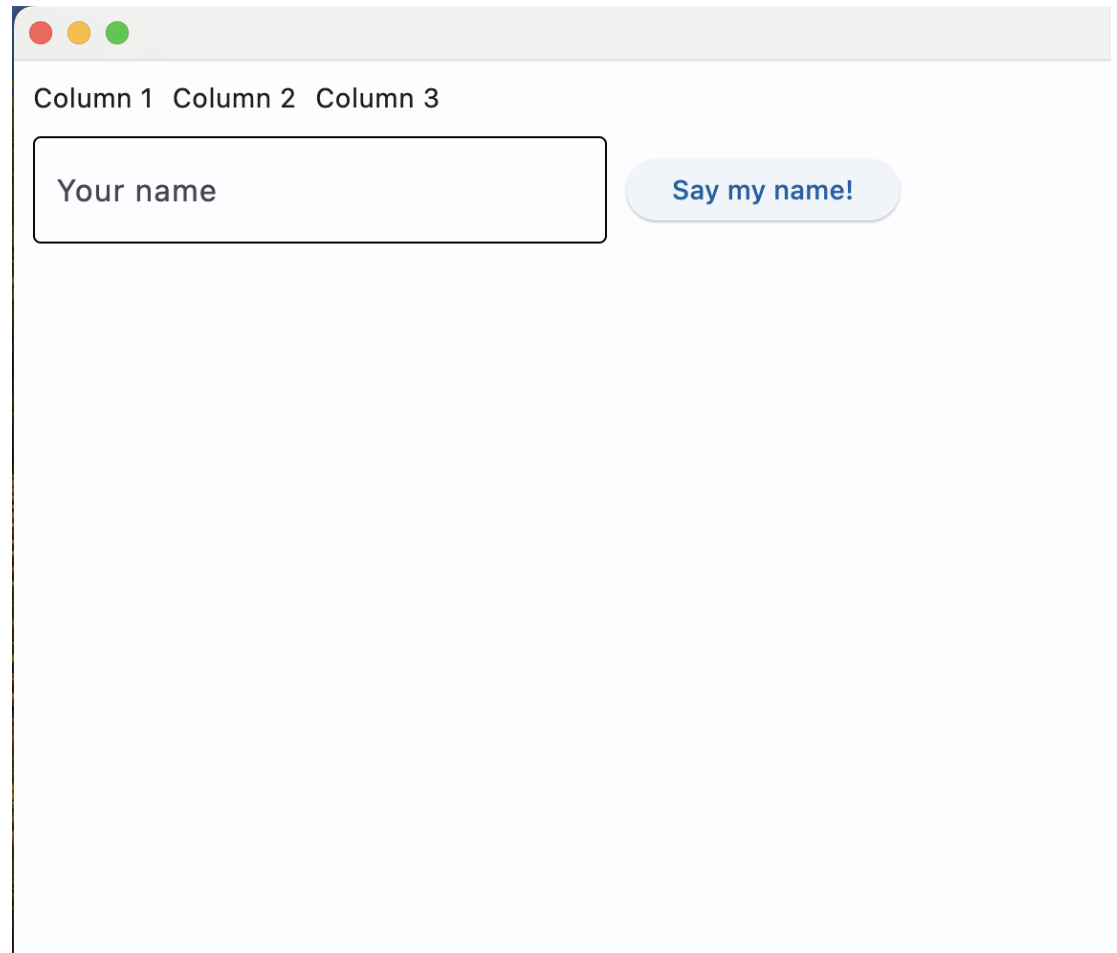
- Elements in Page are updated only when `page.update()` is called
- Note that the content of Controls can be updated anytime, and not only when added to the page, by simply changing the value field

```
1 import flet as ft
2 import time
3
4 def main(page: ft.Page):
5     t = ft.Text()
6     page.add(t) # it's a shortcut for page.controls.append(t) and then page.update()
7
8     for i in range(10):
9         t.value = f"Step {i}"
10        page.update()
11        time.sleep(1)
12
13 ft.app(target=main)
```

Step 8

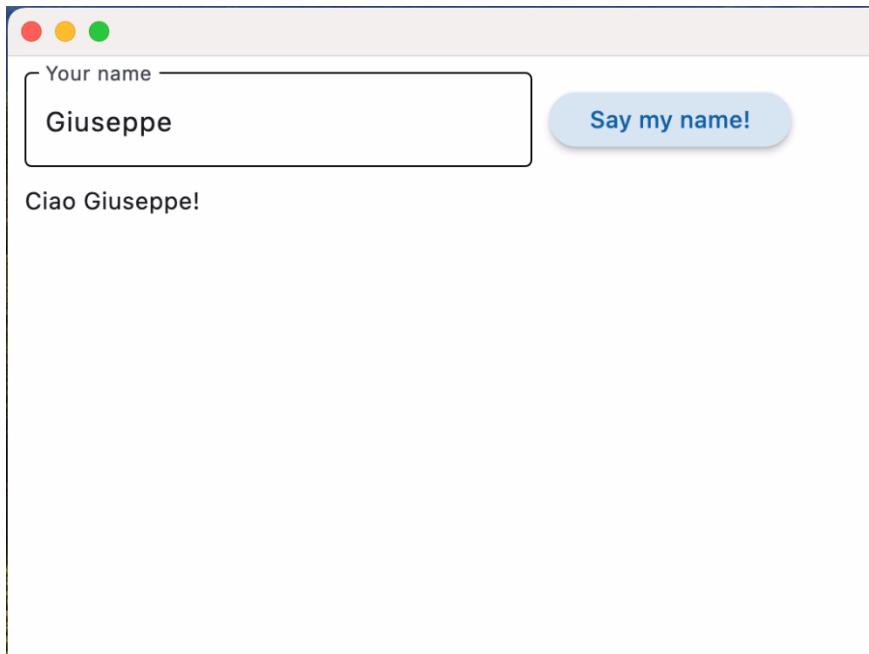
Giving some structure

```
1 import flet as ft
2
3 def main(page: ft.Page):
4     page.add(
5         ft.Row(controls=[
6             ft.Text("Column 1"),
7             ft.Text("Column 2"),
8             ft.Text("Column 3")
9         ])
10    )
11
12    page.add(
13        ft.Row(controls=[
14            ft.TextField(label="Your name"),
15            ft.ElevatedButton(text="Say my name!")
16        ])
17    )
18
19 ft.app(target=main)
```



Event handlers

- Some control items can trigger events:
 - e.g. buttons when clicked



```
1 import flet as ft
2
3 def main(page: ft.Page):
4
5     def handle_button(e):
6         page.add(ft.Text("Ciao " + name.value + "!"))
7
8     name = ft.TextField(label="Your name")
9     page.add(
10         ft.Row(controls=[
11             name,
12             ft.ElevatedButton(text="Say my name!", on_click=handle_button)
13         ])
14     )
15
16 ft.app(target=main)
```



Many options

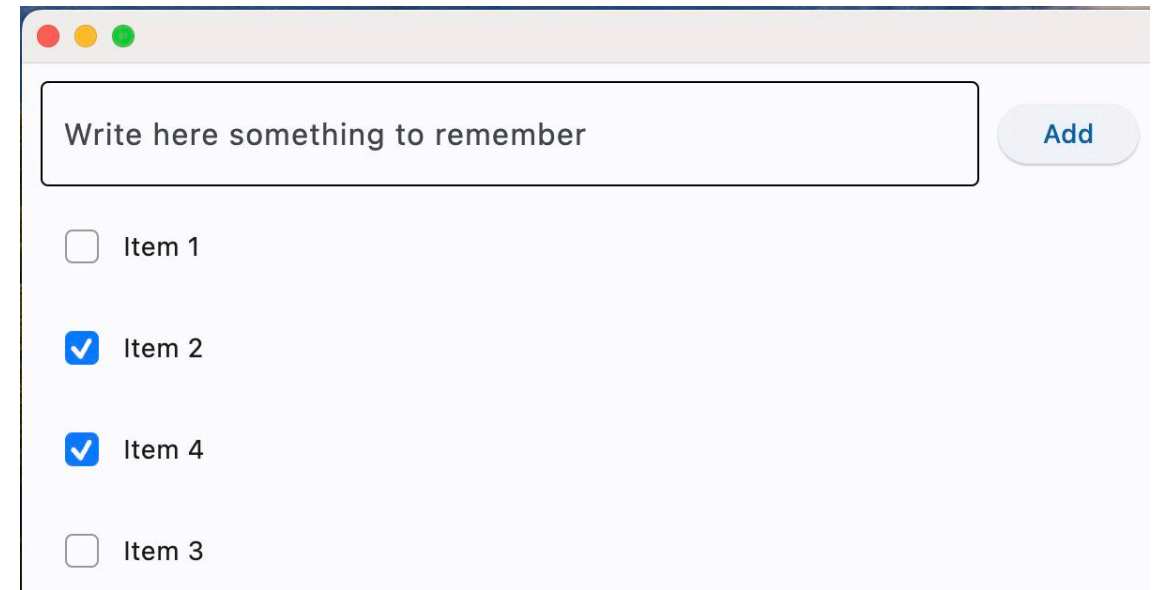
- <https://flet.dev/docs/controls>

tutti i controlli



Example: ToDo list

- Let's try to get something like this

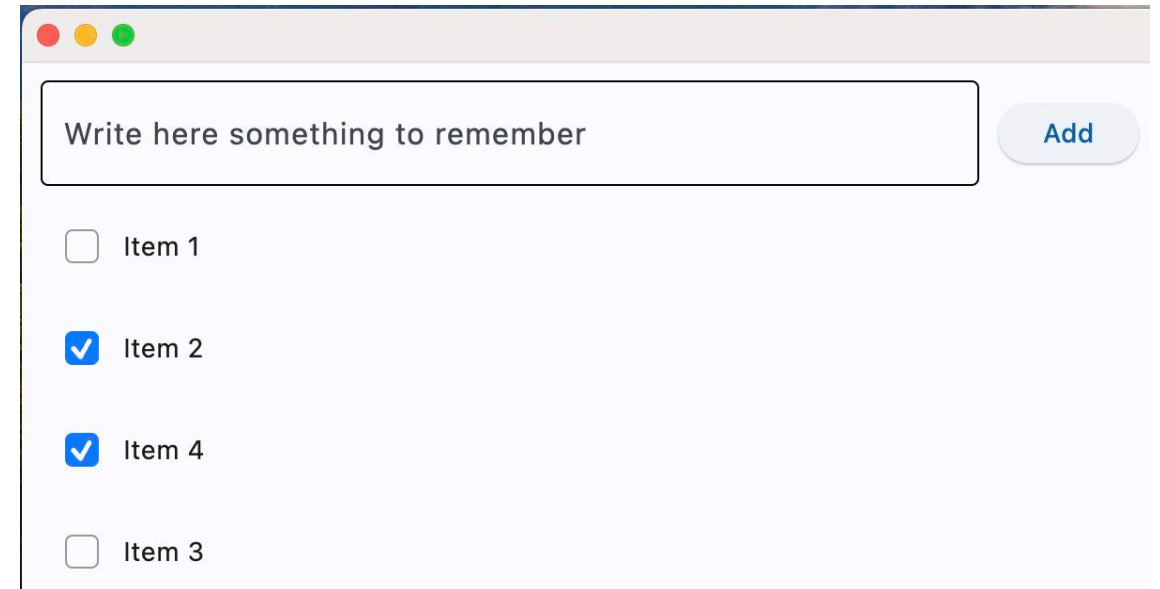


Write here something to remember [Add](#)

- ☐ Item 1
- ☒ Item 2
- ☒ Item 4
- ☐ Item 3

Example: ToDo list

```
1 import flet as ft
2 def main(page):
3
4     def handleAdd(e):
5         page.add(ft.CupertinoCheckbox
6                 (label=txtBox.value, value=False))
7         txtBox.value = ""
8         txtBox.update()
9
10    txtBox = ft.TextField(label="Write here something to remember", width=500)
11    button = ft.ElevatedButton(text="Add", on_click=handleAdd)
12    page.add(ft.Row([txtBox, button]))
13
14 ft.app(target=main)
```

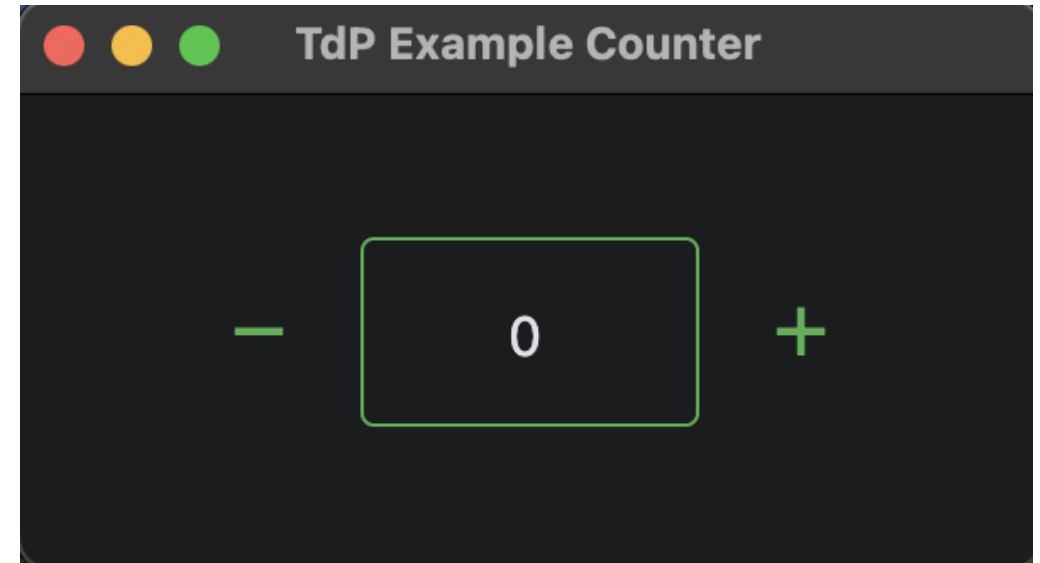


The screenshot shows a web application with a light blue background. At the top, there is a text input field with the placeholder text "Write here something to remember" and a rounded "Add" button to its right. Below the input field, there is a list of four items, each with a checkbox and a label:

- ☐ Item 1
- ☒ Item 2
- ☒ Item 4
- ☐ Item 3

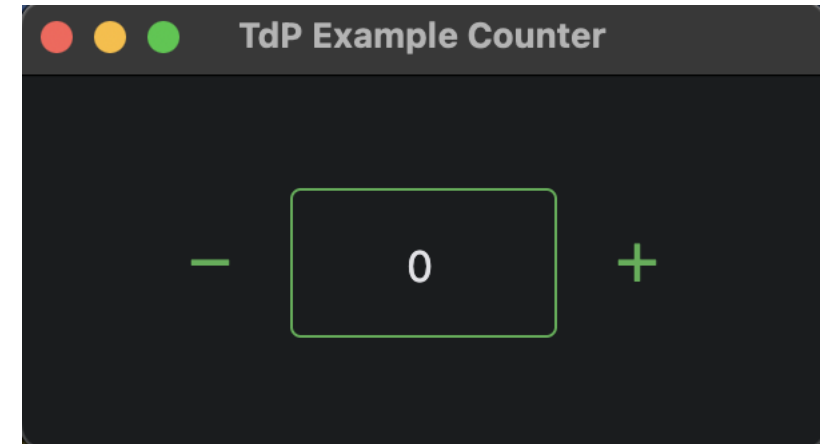
Example: Counter

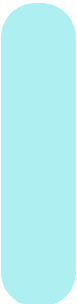
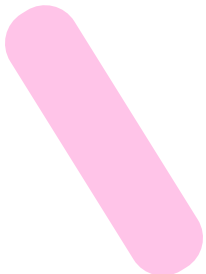
- Let's build a counter



Example: Counter

```
1 import flet as ft
2
3 def main(page: ft.Page):
4     page.title = "TdP Example Counter"
5     page.vertical_alignment=ft.MainAxisAlignment.CENTER
6
7     def handleRemove(e):
8         val.value = val.value-1
9         val.update()
10    def handleAdd(e):
11        val.value = val.value+1
12        val.update()
13
14    val = ft.TextField(width=100,border_color="green",value=0,text_align="center")
15    page.add(ft.Row([
16        ft.IconButton(icon=ft.icons.REMOVE,icon_color="green",on_click=handleRemove),
17        val,
18        ft.IconButton(icon=ft.icons.ADD,icon_color="green",on_click=handleAdd)
19    ],
20    alignment=ft.MainAxisAlignment.CENTER
21    ))
22    page.update()
23
24 ft.app(target=main)
```





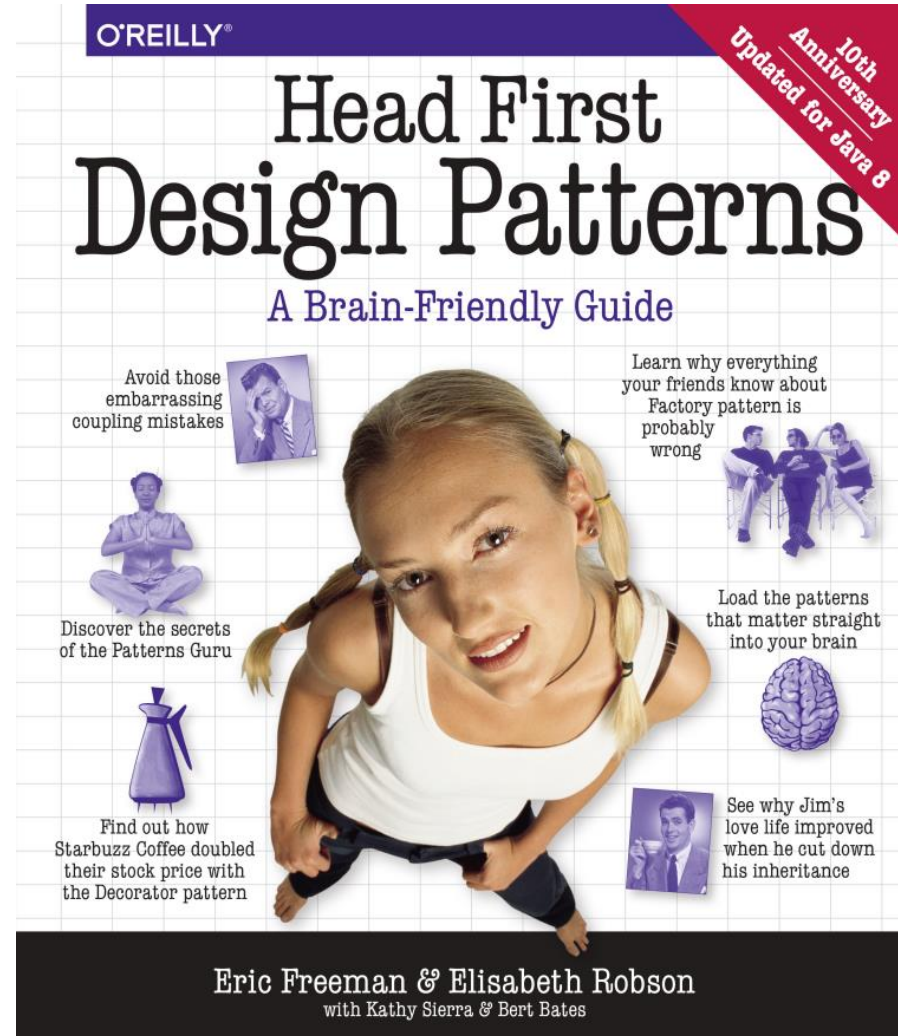
PATTERN MVC

Application complexity and MVC

Model
View
Control

- Interactive, graphical applications exhibit complex interaction patterns
- Flow of control is in the hand of the user
- Actions are mainly asynchronous
- How to organize the program?
- Where to store data?
- How to decouple application logic from interface details?
- How to keep in sync the inner data with the visible interface?

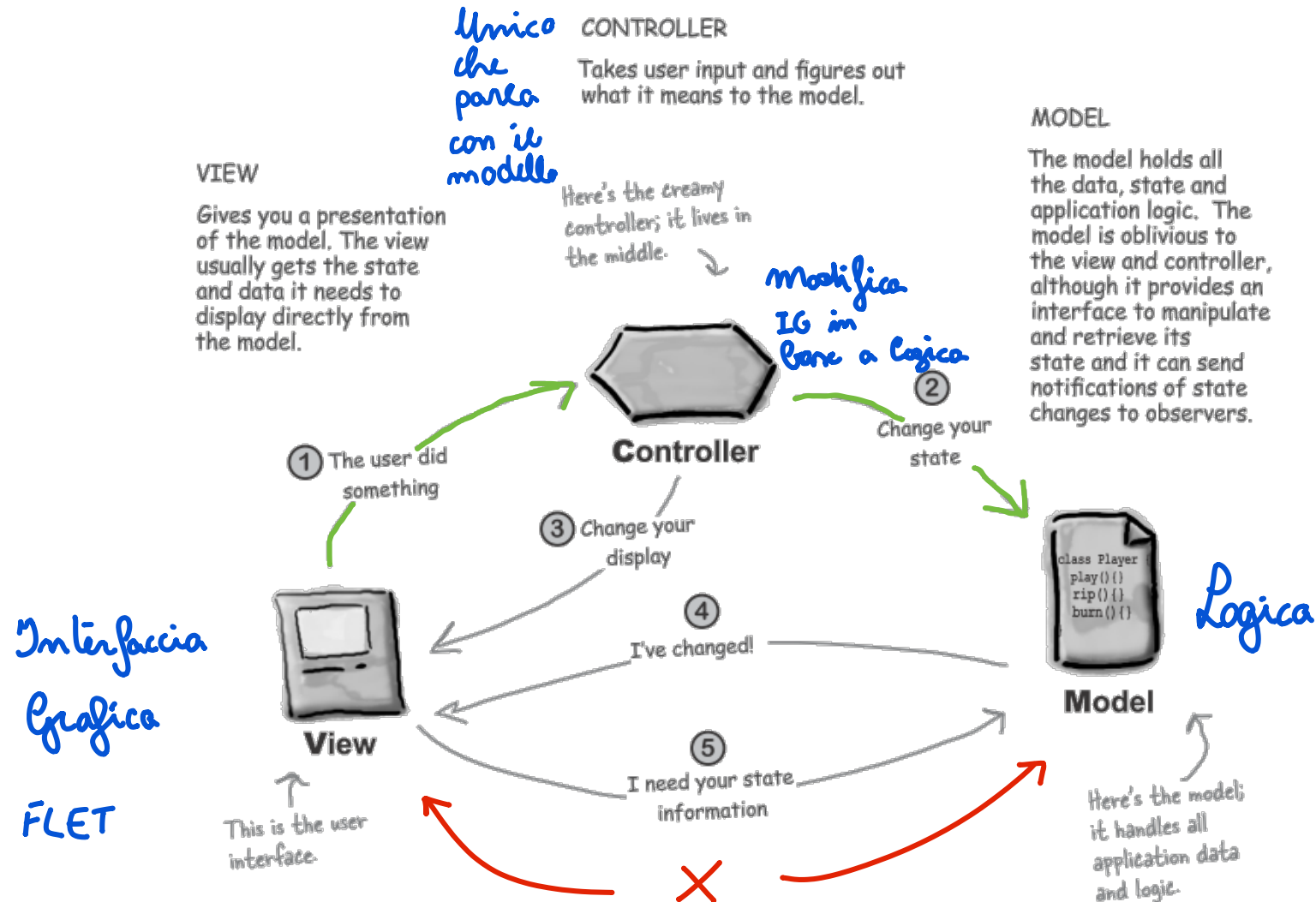
Design Patterns



Design Patterns

- How to build systems with good OO design qualities
 - Reusable, extensible, maintainable
- Patterns: Proven solutions to recurrent problems
 - Design problems
 - Programming problems
- Adopt and combine the OO constructs
 - Interface, inheritance, abstract classes, information hiding, polymorphism, objects, statics, ...
- Help dealing with changes in software
 - Some part of a system is free to vary, independently from the rest

MVC pattern defined



Normal life-cycle of interaction

- ① **You're the user — you interact with the view.**
The view is your window to the model. When you do something to the view (like click the Play button) then the view tells the controller what you did. It's the controller's job to handle that.
- ② **The controller asks the model to change its state.**
The controller takes your actions and interprets them. If you click on a button, it's the controller's job to figure out what that means and how the model should be manipulated based on that action.
- ③ **The controller may also ask the view to change.**
When the controller receives an action from the view, it may need to tell the view to change as a result. For example, the controller could enable or disable certain buttons or menu items in the interface.
- ④ **The model notifies the view when its state has changed.**
When something changes in the model, based either on some action you took (like clicking a button) or some other internal change (like the next song in the playlist has started), the model notifies the view that its state has changed.
- ⑤ **The view asks the model for state.**
The view gets the state it displays directly from the model. For instance, when the model notifies the view that a new song has started playing, the view requests the song name from the model and displays it. The view might also ask the model for state as the result of the controller requesting some change in the view.

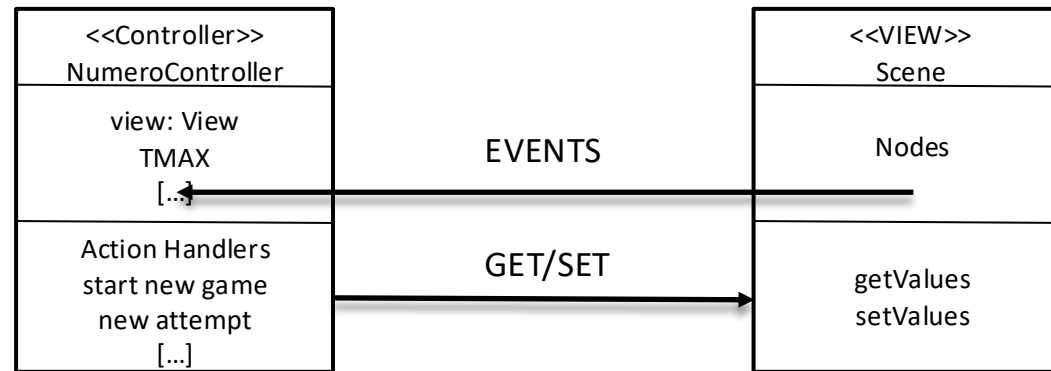
Mapping concepts to Python

- View: presenting the UI
 - View class, which instantiates controllers defined by FLET
 - interacts with the controller
- Controller: reacting to user actions
 - Set of event handlers
 - Local variable to handle the interface status
- Model: handling the data
 - Class(es) including data
 - Persistent data in Data Bases

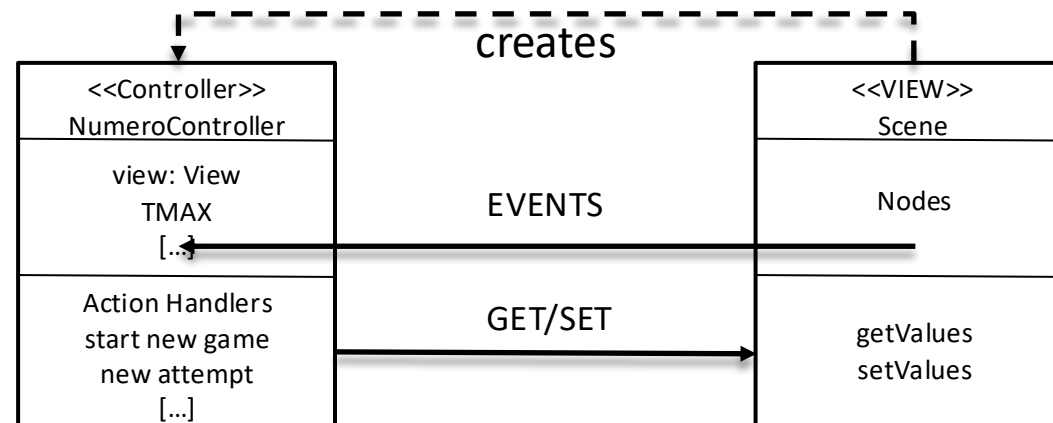
Exercise

- Update the «IndovinaNumero» by using the MVC pattern
- Where do you declare the data class?
- Which class should have access to which?
- Who creates what objects?

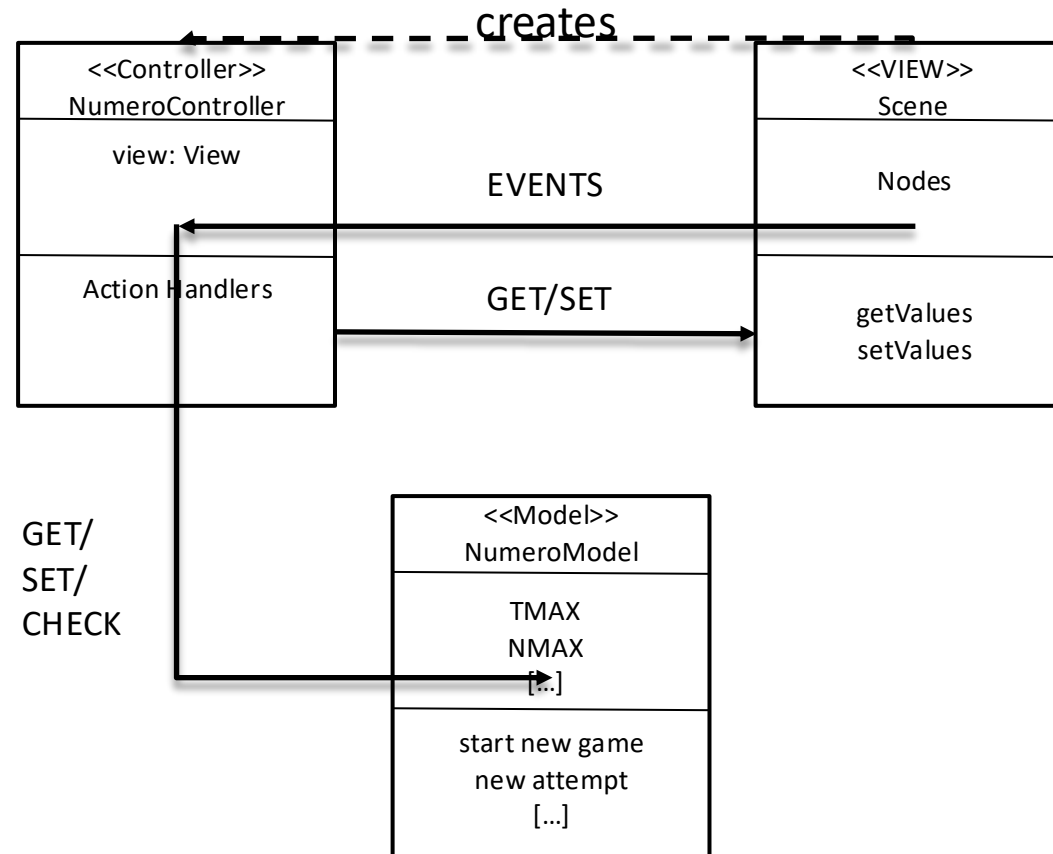
A possible solution



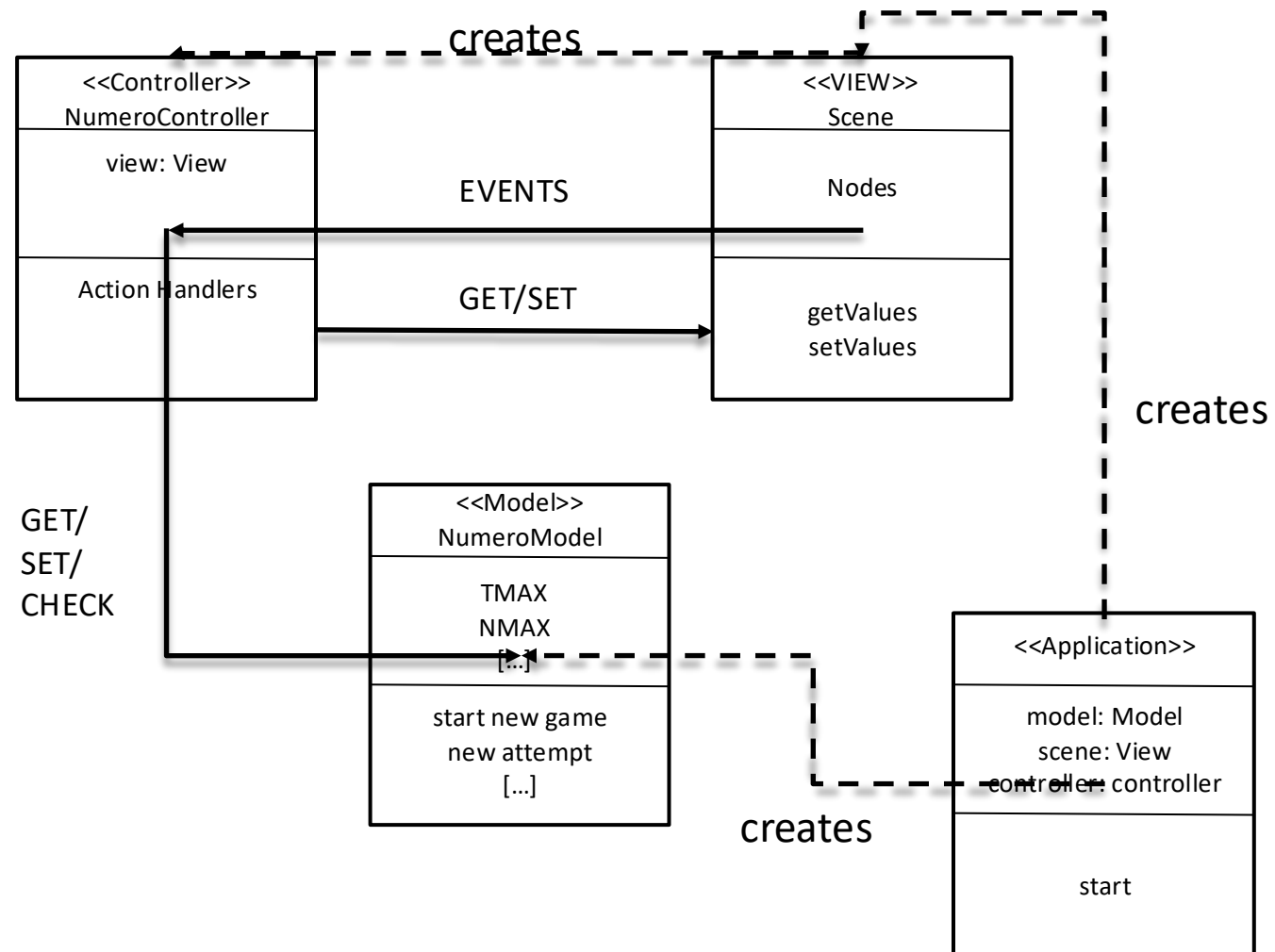
A possible solution



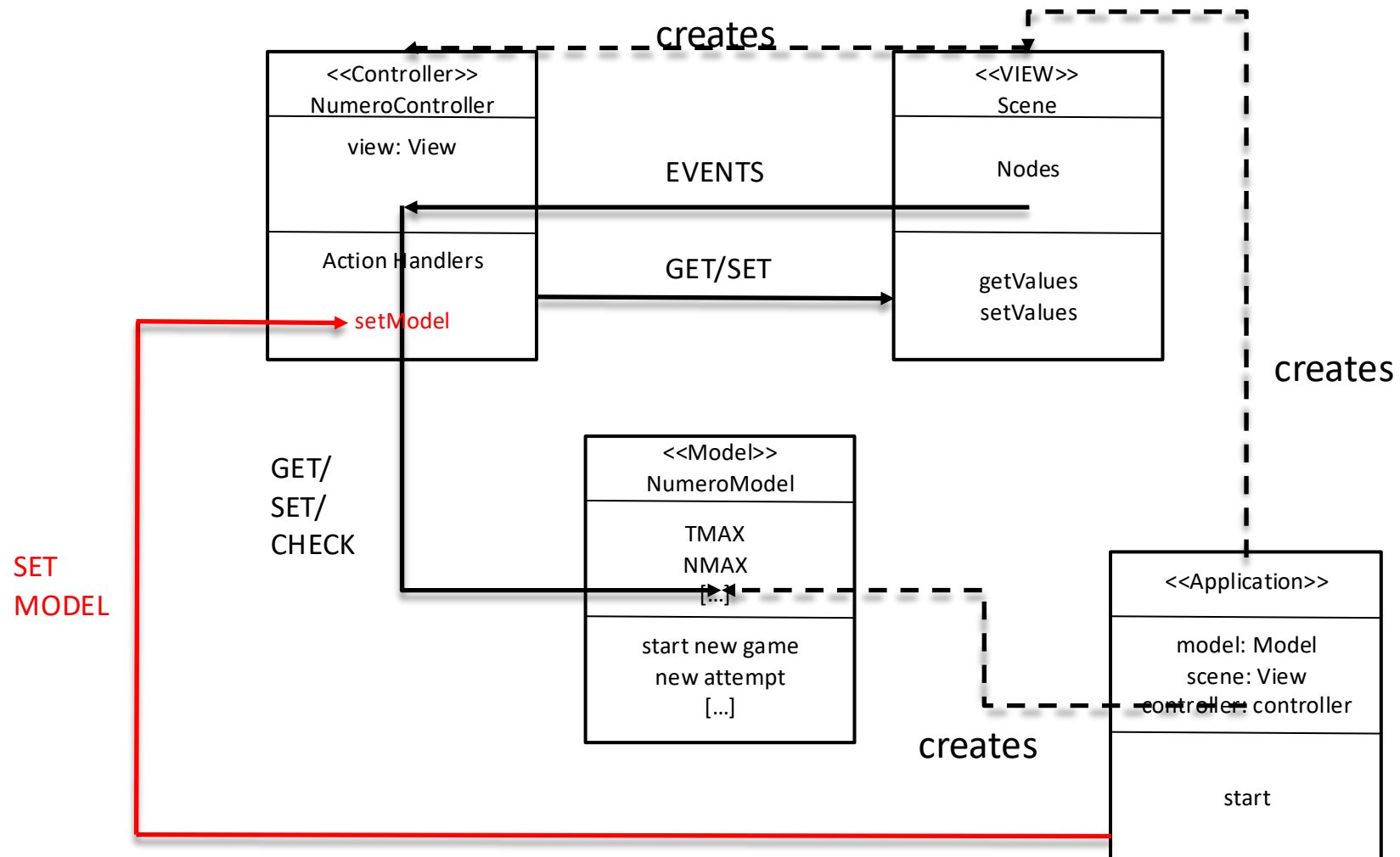
A possible solution



A possible solution



A possible solution



Example: Libretto

TdP with FLET (2024-2025)

☐ Light theme

Il mio Libretto Voti

Lista degli esami finora sostenuti:

- a: 24
- b: 21
- c: 26
- d: 28
- e: 27
- f: 23
- g: 26
- h: 19
- i: 21
- j: 28

Example: Indovina il Numero

Indovina il Numero - edizione TdP 2024

Numero Max
100

Tentativi Rimanenti
2

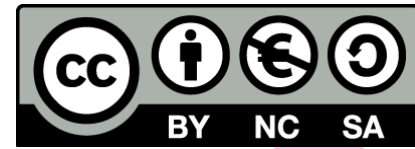
Tentativi Max
6

Valore

Indovina

Indovina a quale numero sto pensando:
Nope! il numero segreto è più grande di 50.
Nope! il numero segreto è più grande di 75.
Nope! il numero segreto è più piccolo di 88.
Grande! il numero segreto era 80.

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